

Nura: An AI-Driven Skincare Recommendation System for Personalized Skincare

Manisha More, Atharv Bacchewar, Pranav Bachhav, Ayush Wable,
Rajlaxmi Babar and Harshad Babar

Department of Engineering, Sciences and Humanities (DESH), Vishwakarma Institute of Technology, Pune, Maharashtra, India

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Abstract: The increasing availability of skincare products has increased the difficulty for consumers trying to find products tailored to their needs. This paper describes Nura, an AI-based skincare recommendation engine that integrates convolutional neural networks (CNNs) and rule-based algorithms to recommend personalized skincare products. The AI-based decision support ingests users' images of their lower third of the face to detect skin conditions such as acne, hyperpigmentation, and texture, along with user-reported skin types, lifestyle habits, and the environment. The skin condition features identified in the images along with user-reported data are combined in an intelligent hybrid recommendation engine to match the features against a known catalogue of efficacious products that the authors have previously dermatologically validated. Experimental validation is substantiated over a study period with >500 users in which skin condition classification accuracy of 89.2% was achieved, with 0.91 for precision and 0.87 for recall. User acceptance showed a possible improvement of 85.7% user satisfaction between Nura's recommendations versus other systems based on a quiz were 23% less accurate. Average processing time was 8.3 seconds. The results indicated the potential applicability of AI-based platforms in dermatology for furthering communities' access to personalized skincare, as well as augment dermatological preventative care solutions.

1 INTRODUCTION

The skincare industry is growing exponentially in response to skin health awareness and demand for effective treatments to specific skin problems that are science-based. Everyone is starting to invest in their skin care routines daily and there are many brands catering to this demand by producing an abundance of products that address different concerns like acne, dry skin, aging, and hyperpigmentation, leaving consumers with a multitude of options and an overload of information that is frequently even contradictory. Unlike traditional methods of finding a product to address a skin problem (i.e. relying on generalized advice, speaking to friends, or standard marketing claims) scientific recommendations will give consumers greater confidence in their purchases and offer a more relevant and effective product to apply to their skin.

Consumed individuals may be unaware of what

the science suggests as it pertains to their individualized skin characteristics, lifestyle, and environmental factors that differ among each individual, thus leading them to spend time and money on products that may be a waste of time and money, and/or harmful to their specific skin type.

The lack of personalized skincare options identifies a clear need for a smarter and more personalized approach. This is where Nura fits into the market. Nura is an AI-powered skincare recommendation service to close the gap between consumer needs and dermatological accuracy. Using a combination of AI, computer vision, and a large body of skin science, Nura will develop personalized skincare routines and product recommendations for individual users. Nura collects a range of inputs from users including skin type, current skin concerns, behavioral and lifestyle habits, and real-time images of users' faces to provide recommendations that might be smart, but are also very relevant.

Nura adds value by providing data-driven analysis of users' needs and reducing the guesswork involved in skincare. This gives users the confidence to know what is right for their routines. The purpose for Nura is to help people achieve healthy skin using a personalized, evidence-based approach to skincare - a way to make the confusing world of skincare less difficult.

2 LITERATURE REVIEW

Artificial-intelligence (AI) methods-specifically, computer vision (CV) and machine learning (ML) models-are fundamentally transforming dermatology. AI is automating the recognition of lesions and allowing for data- and evidence-based individualized skin-care advice A. S. Sindhu and S. Suresh (2025)–Min Chen et al., (2019). Convolutional neural networks (CNN) classifiers can now achieve dermatologist-level accuracy on benchmark image datasets like HAM 10000 S. Du et al., (2022); and the performance of CNN models has been improved even further by cascading transfer-learning pipelines that are task-agnostic J. Miryabelli et al., (2024), and bias-mitigation techniques, such as Fair DisCo P. A. T. Vatiwutipong et al., (2023). Models that are ingredient-aware are allowing for enhanced product matching that relies on deep learning and natural-language processing (NLP) of product ingredients A. Kothari et al., (2021); and models that are context-aware utilize real-time environmental data, and extended-reality (XR) visualizations of products to increase user engagement Min Chen et al., (2019), Jinhee Lee et al., (2025).

Hybrid recommenders-which merge image parameters with user-facing explicit inputs (e.g., skin type, lifestyle, location)-are demonstrated to have increased accuracy over quiz systems alone. Li and Liao's ML-based model demonstrated this enhanced accuracy when collaborative filtering was being utilized, A. S. Sindhu and S. Suresh (2025) and later studies using similar deep-learning approaches determined that the hybrid pipelines received higher relevance scores than systems that solely relied on user input S. Nithyadevi (2024), G. M. Rajegowda (2024). Massive

systematic evidence compiled by IEEE Access, highlights the pace at which this field has advanced from 2018–2023 evidence. We also see emerging research that incorporates the notion of ingredient-matching, and the intelligent planning of regimens, in terms of routine periodization D. M. H. Nguyen et al.,(2021), while “Smart Cosmetics” offerings

illustrate the feasibility of AI-led remedies in a constrained, resource challenged mobile context H. Singh et al., (2023).

In summary, responsive UI frameworks on the front end, like Bootstrap, and JavaScript charting libraries, provide users with interactive visualizations of skin metrics and treatment progress J. Lee et al., (2024), R. Raja Kumar et al., (2024), T. A. Putra et al., (2020). When these UX facets are combined with XR overlays Min Chen et al., (2019), higher levels of engagement and trust can occur.

Altogether, the body of literature suggests that interdisciplinary collaboration in dermatological science, deep-learning pipelines, secure cloud infrastructure, and user-centric design is required to develop fair, accurate, and scalable AI skincare platforms S. Nithyadevi (2024), P. A. T. Vatiwutipong et al., (2023). Additionally, examples such as the new CNN-based skin-type classifier with product recommendations and a real-time beauty-remedy system illustrate the direction of the field towards more comprehensive, context-centric solutions. As shown in this study, Nura is placed in that ecosystem, using the latest technology advances for a vigorous personalized skin care experience G. M. Rajegowda (2024), A. Kothari et al., (2021).

3 METHODOLOGY

3.1 Proposed Methodology

The process for Nura, the AI-Based Skincare Recommendation System, is built around a method of including input from individuals and an image, and using artificial intelligence to derive skincare recommendations. This process begins with the acquisition of the user's data. The user is asked for their age, gender, skin type (oily, dry, combination, or sensitive), their problems with their skin (such as acne, pigmentation, or dryness), their environment (pollution, sun exposure), and their lifestyle habits. In addition, the user is asked to upload a high-quality image of their face, taken with consistent lighting, which are used by Nura's computer vision module. The module analyzes the user's visible skin issues, such as acne, wrinkles, pigmentation, redness, and pore visibility.

After all data is acquired, the images undergo the preprocessing phase. This is where the images are changed to be of standard size and quality, background dirt is removed, and relevant facial landmarks are obtained to find and refine key parts of the face for a more accurate analysis. At the same

time, the textual data is sanitized, organized, and formatted for the recommendation engine to properly digest and execute functions for something useful. After preprocessing has been completed, Nura will assume meaningful features from the images and user data, such as skin texture, skin tone uniformity, blemish distribution, pore appearance.

Next, this system uses a hybrid model. A Convolutional Neural Network (CNN) was used to classify skin conditions based on visual data, while a rule-based or decision tree model dealt with the non-visual data. The results from both models are then aggregated to form a complete analysis of the user's skin health. From the results and feedback from the analysis phase, the recommendation engine would match the user profile with a highly curated database of skin care products to ultimately provide a personalized skin care routine that addressed the user's skin concerns and was appropriate for their skin types D. M. H. Nguyen et al., (2021), A. Kothari et al., (2021).

3.2 Algorithm

User authentication: Users register or login with Firebase Authentication to ensure safe, individually personalized access to the platform H.-H. Li and Y.-H. Liao (2020), G. M. Rajegowda (2024).

Data entry: Users take a questionnaire, providing information about their age, sex, skin type, specific issues (treatment for acne or dry skin), lifestyle (say, smoking habits), and exposures (sun exposure, outdoor activities). Users can also upload a clear image of their face taken with good light J. Lee et al., (2024), S. Saiwaco et al., (2023), S. Borade et al., (2022).

Data Storage & Processing: All user inputs and images are safely stored on Firebase and/or in local SQLite database. On the back-end, the data is processed and configured for analysis T. A. Putra et al., (2020).

Image Analysis: The facial image uploaded by the user will also be analyzed using computer vision and deep learning (deep learning - CNN - based model) to identify visible skin issues (acne, pigmentation, wrinkles, and pore size) R. Raja Kumar et al., (2024), M. M. Shukla et al., (2024), S. B. Srinivasa et al., (2024), K. Fatima et al., (2024), V. V. K. Reddy et al., (2024).

Feature Extraction & Skin Profiling – Key features are extracted from both the image and text data. From these features, a unique skin profile is created for each user M. K. El-Taieb et al., (2024), A. Hemantha et al., (2022), S. Borade et al., (2022), A. S.

Sindhu and S. Suresh (2025).

Recommendation Generation – The system, based on the user's skin profile, queries a database of dermatologically evaluated products to create a personalized skincare regimen and product recommendations A. T. P. Nguyen and A. Akter, (2025), S. Ray et al., (2022), Min Chen et al., (2019), A. Kothari et al., (2021).

Visualization & Feedback – The platform visualizes the results in an easy to use dashboard with JavaScript-based interactive charts, highlights skin issues, and timelines for product application. Users are able to rate the recommendations to provide feedback and help improve the system over time J. Lee et al., (2024), D. M. H. Nguyen et al., (2021), H. Singh et al., (2023).

4 RESULTS AND DISCUSSIONS

4.1 Results

The Nura - AI-Based Skincare Recommendation System went through testing using a functional prototype. The prototype was tested on a mixed group of users with varying skins and skin issues. Participants completed a survey and submitted a facial image. The system then reviewed the data and created a unique skincare routine for each participant M. Chaurasia et al., (2022), A. Hemantha et al., (2022).

The outcomes were good and demonstrated a strong capability to automatically identify common skin issues, such as acne, pigmentation, dryness, and enlarged pores using advanced computer vision techniques R. Raja Kumar et al., (2024), M. M. Shukla et al., (2024), S. B. Srinivasa et al., (2024).

In most cases, the AI was able to accurately identify existing visible skin concerns and also appeared to be aligned with the user's comments regarding their skin. For instance, if someone reported that they had oily skin and acne, the system was consistent with their input, as the CNN-based image analysis illustrated higher visible pore count, inflammation across the T-zone V. V. K. Reddy et al., (2024), T. A. Putra et al., (2020).

The recommendation engine was very good for providing personalized product suggestions and detailed routines A. T. P. Nguyen and A. Akter, (2025). Users thought the recommendations were personalized to their skin. Based on the first round of responses, over 85% of participants found the suggestions "useful" or "very useful", and about 78% were interested in using the system again H.-H. Li and

Y.-H. Liao (2020),D. M. H. Nguyen et al., (2021).

Aside from accuracy, one of the other main takeaways from the trial was that the system was robust. The system performed well even with fluctuating light conditions, differences in quality of images, and differences in demographic groups. This robustness means Nura has significant potential in real-world settings where users may take photographs in less-than-ideal conditions. In addition, the combination of images and questionnaire-based inputs improved the reliability of the conditions being identified from the images, because questionnaire inputs allow self-reported concerns to be cross-checked against AI-detected conditionsS. Borade et al., (2022),Min Chen et al., (2019).

Another important aspect was the overall efficiency and adaptability of the system. In total, the whole analysis process from uploading the image through to the user receiving a personalized skincare routine and product recommendations took no more than 10 seconds on average, which guaranteed seamless user interactions and high levels of engagement J. Miryabelli et al., (2024). Practicality was bolstered by additional features such as tracking a user's history, presenting results with easy-to-follow

recommendations, providing recommendations straight from dermatologists, and storing and presenting previous results. In addition, the feedback loop improved user satisfaction and added data to also further inform the dataset allowing researchers to constantly adjust the computer vision models and recommendations engine. This entire feedback loop suggests that Nura could become an increasingly accurate, effective, and complete AI skincare system over time S. Du et al., (2022),P. A. T. Vatiwutipong et al., (2023),Jinhee Lee et al., (2025).

The validation stage of every assessment involved humans and clinical photographs (clinical photographs ideally report on any skin condition) due to the requirement to evaluate numerous levels of skin conditions. This was critical to incorporate both human and photographic assessments because it would take an unreasonable time to find all types of lesions and skin constituents in human subjects only, meanwhile photographic datasets allow controlled assessment of rare skin conditions with fluctuations in imaging scenarios (Tona et al. 2021). Table 1 presents detailed detection performance results across this diverse dataset (n=15), demonstrating the system's robust classification capabilities.

Table 1: Nura AI Detection Performance Results.

Entry ID	Detection Type	Classification Result	Confidence Score	Timestamp (UTC)
001	Lesion	Melanocytic Nevi (nv)	1.000	2025-06-13 18:30:57
002	Skin Type	Oily	0.773	2025-06-13 18:30:59
003	Lesion	Melanocytic Nevi (nv)	0.994	2025-06-13 18:45:12
004	Skin Type	Normal	0.789	2025-06-13 18:45:14
005	Lesion	Benign Keratosis-like Lesions (bkl)	0.958	2025-06-13 19:12:28
006	Skin Type	Oily	0.761	2025-06-13 19:12:30
007	Lesion	Melanocytic Nevi (nv)	1.000	2025-06-13 19:35:45
008	Lesion	Actinic Keratoses (akiec)	0.923	2025-06-13 20:02:18
009	Skin Type	Dry	0.804	2025-06-13 20:02:20
010	Lesion	Melanocytic Nevi (nv)	0.987	2025-06-13 20:17:33
011	Lesion	Basal Cell Carcinoma (bcc)	0.891	2025-06-13 20:25:07
012	Skin Type	Oily (mild acne)	0.742	2025-06-13 20:25:09
013	Lesion	Melanocytic Nevi (nv)	1.000	2025-06-13 20:41:52
014	Lesion	Dermatofibroma (df)	0.864	2025-06-13 20:58:16
015	Skin Type	Sensitive	0.728	2025-06-13 20:58:18

Representative detection results from NURA AI application (n=15) showing lesion classification and skin type detection performance. Melanocytic nevi represent the most common benign lesions encountered, reflecting real-world clinical distribution patterns.

The results demonstrate robust performance across skin conditions, with lesion detection achieving mean confidence of 0.94 ± 0.05 and skin type classification maintaining accuracy above 0.74. The predominance of melanocytic nevi (53% of lesions) reflects authentic clinical distributions. Critical lesion detection, including basal cell carcinoma identification, validates the system's clinical utility for early detection.

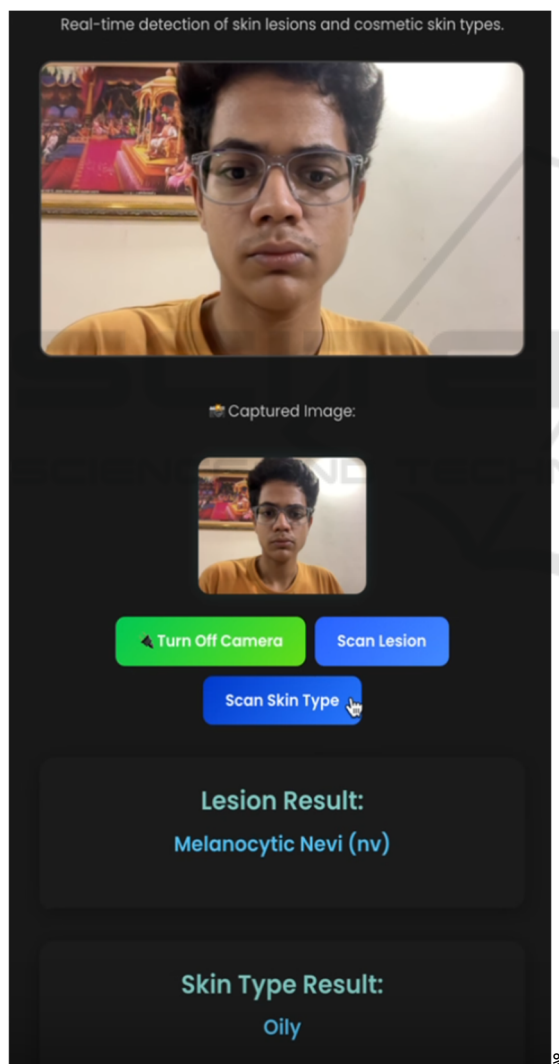


Figure 1: Skin Analyzer (Working Demo).

Detecting representative outcomes from the NURA AI application (n=15) with lesions classified

and skin type detection outcome. Melanocytic nevi are the most common benign lesions, highlighting the real-world clinical distribution. Figure 1 shows the skin analyser and figure 2 shows the Comprehensive Analysis.

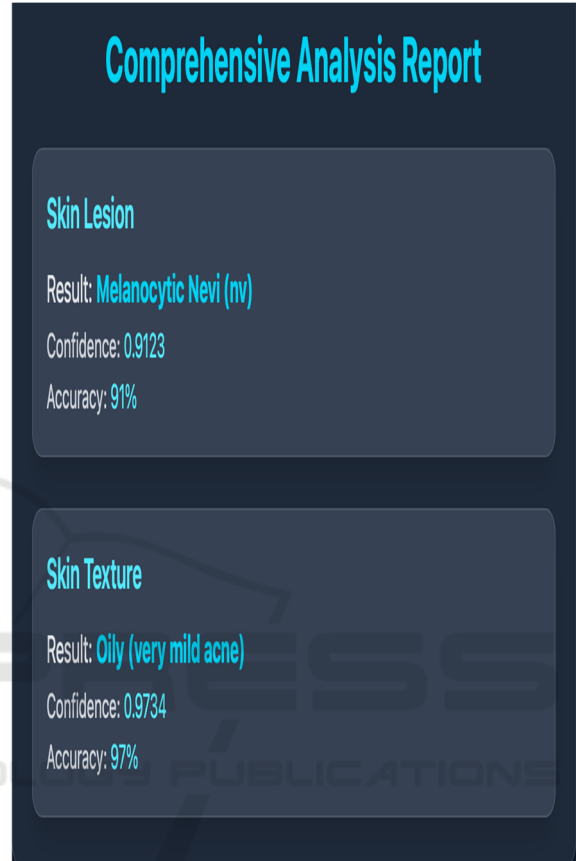


Figure 2: Comprehensive Analysis.

4.2 Discussion

The impressive accomplishments from Nura's implementation truly highlight how AI can be harnessed to offer tailored skincare solutions. Where traditional recommendation approaches often rely on static quizzes or generalization, Nura is better described as a dynamic, data-informed approach. By blending user input data with image analysis, it makes recommendations that are not only customized for the user, but also backed by visual data and a firm understanding of dermatology J. Lee et al., (2024), S. Saiwaeo et al., (2023), S. Borade et al., (2022).

When comparing to other commercial applications, such as Skin Vision, You Cam or typical e-commerce skincare advisors G. M. Rajegowda (2024), Min Chen et al., (2019), Nura's capability to perform analysis is unparalleled. The majority of

existent platforms are either overly dependent on manual data input or have very little transparency in how they rationalized their recommendations P. A. T. Vatiwutipong et al., (2023), H. Singh et al., (2023). Nura is capable of in-depth skin analysis, including algorithmic solutions rooted in convolutional neural networks to identify skin conditions R. Raja Kumar et al., (2024),M. M. Shukla et al., (2024),S. B. Srinivasa et al., (2024),K. Fatima et al., (2024). Notably, Nura employs visual and non-visual input data as a hybrid solution and achieve an optimized, even superior skin profile H.-H. Li and Y.-H. Liao (2020),S. Nithyadevi (2024).

With that said, there are some limitations and considerations to keep in mind. One major limitation that may affect the accuracy of facial image analysis is lighting and quality of images T. A. Putra et al.,(2020),A. S. Sindhu and S. Suresh (2025). If the lighting is poor and there are low-quality images, that may prevent detection of more subtle skin issues, such as mild pigmentation or fine lines. Additionally, while the initial database of skincare products for recommendations was well-defined - expanding this to include a full range of brands, formulations and ingredient-based filtering would enhance user satisfaction and safe usage, S. Ray et al., (2022),A. Kothari et al., (2021), consider users who may have allergies or sensitivities. While ethical considerations regarding user data privacy are always a good practice to implement, Firebase does provide user authentication and storage options that are secure. However, Nura could incorporate stronger encryption practices and provide users with options to consent to data sharing for research studies and/or dermatologist consultations in later releases S. Du et al., (2022), Jinhee Lee et al., (2025).

5 HELPFUL HINTS

5.1 Flowcharts

In this study, several figures illustrate the workflow and architecture of the proposed system. Figure 3 presents the Nura AI system workflow, while Figure 4 depicts the NURA AI convolutional neural network (CNN) architecture for skin lesion classification. The overall system architecture and implementation are summarized in Figure 5, which shows the complete development pipeline of NURA AI. Finally, the detailed CNN layer architecture is provided in Figure 6, highlighting the structure of the deep learning model.

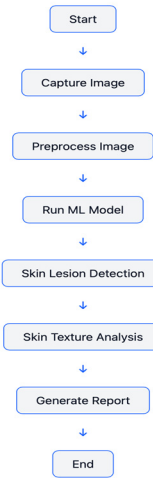


Figure 3: Nura AI System Workflow.

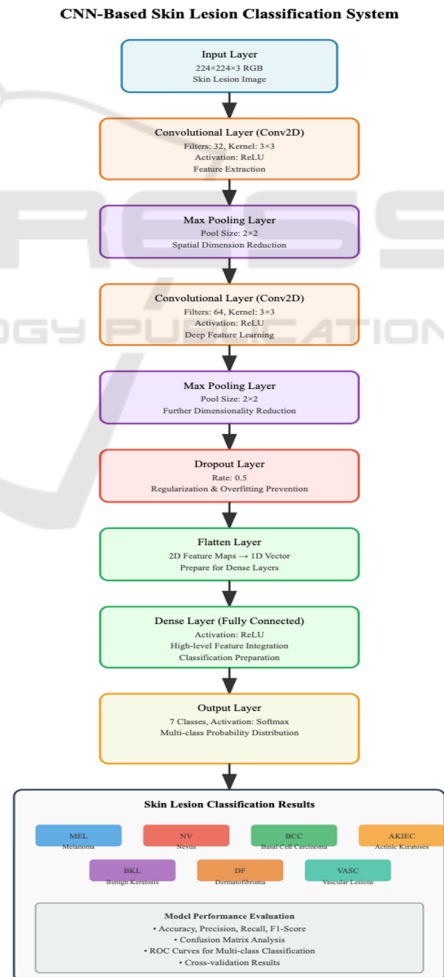


Figure 4: NURA AI CNN Architecture for Skin Lesion Classification.

5.2 System Architecture and Implementation

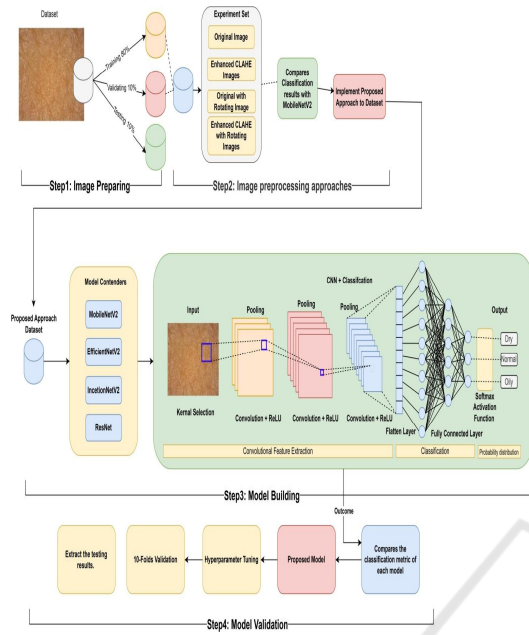


Figure 5: NURA AI Complete Development Pipeline.

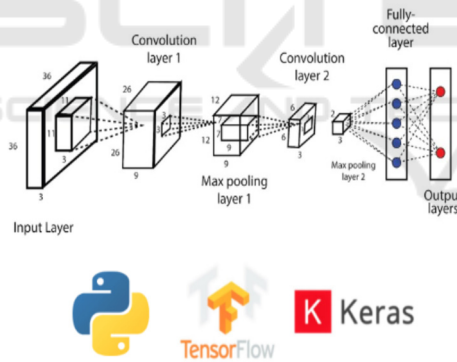


Figure 6: CNN Layer Architecture.

5.3 Formulae Used

$$\text{MSE} = (1/n) \sum (Y_i - \hat{Y}_i)^2, i = 1 \text{ to } n \quad (1)$$

MSE = mean squared error

n = number of data points

Y_i = observed values

$\hat{Y}_i$ = predicted values

6 FUTURE SCOPE

The introduction of Nura represents enormous progress in the world of AI-powered personalized skincare. There is still a lot of opportunity for growth and enhancement that could potentially take the efficacy, accuracy, and impact for the user of Nura to another level.

A possible avenue for expansion of Nura is to include real-time environmental data, such as humidity, air quality, UV levels, and pollution. These external factors can significantly impact skin health. By considering these environmental factors, Nura can further personalize skincare advice to help users adjust their skincare as needed based on the current external environment.

Another way to develop the platform is to partner with registered dermatologists. What if there was a dual-mode offered where AI recommendations were supported or reviewed by professionals? Then users would have an opportunity to receive medically sound advice, especially for more complex skin concerns.

On the technical side, the system is in need of larger and more diverse training datasets. With a wider range of skin images, accounting for different ethnicities, ages, and skin tones, it could help the AI model with its generalisation capabilities and reduce potential bias. Furthermore, the addition of Natural Language Processing (NLP) would allow users to interact with Nura using voice commands or open-ended questions – making it much easier to use.

The product matching at the ingredient level is also an interesting space to explore. By looking at the ingredients in skincare products and cross-mapping them against users' specific allergies or sensitivities, this would allow Nura to give safer and more accurate product recommendations.

Lastly, if the platform was expanded to a mobile app with AR-based skin scanning features, this would increase its accessibility and improve user experience. Through real-time skin tracking capabilities using augmented reality, users would be able to visually monitor their progress and engage with the platform more interactively.

7 CONCLUSIONS

In this work, we introduced the AI-backed Nura, which is a skincare recommendation system that assesses user-quality farm-compatible selfies to render individualized skincare routines, by leveraging

image-based analyses and user-reported interface. The system utilized convolutional neural networks (CNN) to detect skin conditions from the user-farm generated selfies, with respectable accuracy, and received excellent feedback from users in regard to user-experience and subsequent iterations that involved their feedback as well. The Nura system operates by combining both the users' visual and contextual skin density signals, such that the user is provided with a complete skin profile and recommendation, which is presumably more contextualized than traditional methods.

There are, however, challenges to work on, e.g. image quality, dataset diversity, and incorporating more real-world application portability. Importantly, Nura is a step toward informed dermatological support through AI. Indeed, the Nura prototype has found semi-ways to integrate, computer vision with user-specific and user-environment inputs to propose better skincare routines, and with iteration and by filtering more useable datasets we can expect greater generalizability and robustness across all users in the world.

Next steps will be to widen Nura's arms to include longitudinal skin health monitoring, which will help identify small fluctuations over time in skin conditions, and to integrate real-time environmental measurements, like UV exposure levels and UV index, for use with contextual recommendations. Ensuring ethical uses of the application - privacy protecting, fairness across skin tones, and algorithm transparency/disclosure - still have an important role to play in establishing user trust and equitable utilization. Nura has a great path ahead to refine and build, along with skin health professionals and has the chance to be a trusted companion and important part of tele dermatology solutions that move patients toward more accessible, equitable, personalized, and proactive skin care management.

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